

REVISITING PROJECTOR-SIDE POSITIONAL ENCODING IN HIGH-RESOLUTION VISION-LANGUAGE MODELS

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ABSTRACT

High-resolution vision-language models (VLMs) often add a spatial positional encoding (PE) at the multimodal projector even when the upstream vision path already preserves position. We audit this term on VILA-HD-8B using five projector conditions plus an initialization control, matched 312-step cosine schedules, three seeds, and ten benchmarks. Removing the projector PE yields $76.879 \pm 0.323\%$. The learned grid, its zero-initialized control, and uniform-polar Fourier PE yield $76.847 \pm 0.123\%$, $76.899 \pm 0.181\%$, and $77.020 \pm 0.072\%$, respectively. The latter uses 0.46M rather than 2.61M PE parameters and needs no grid interpolation. Separately, changing the learning-rate schedule horizon from 1263 to 312 steps raises the same Cartesian-Fourier model by 3.59 points on five OCR benchmarks. These results show that upstream positional context and matched optimization protocol are essential controls, while specialized projector PE choices have small aggregate differences in this setting.

Index Terms— positional encoding, VLM projector, Fourier features, ablation study, reproducibility

1. INTRODUCTION

High-resolution VLMs use multi-scale processing, tiling, or dynamic patch selection to preserve fine visual detail [1, 2, 3]. A vision encoder maps selected image regions to visual tokens, and a multimodal projector maps these tokens into the language model’s embedding space. Positional information may enter at several stages: inside the vision encoder, during selection-aware resampling, at the projector, or through language-model position IDs [1, 6, 10]. Consequently, evaluating positional encoding (PE) requires specifying both the intervention point and the positional information retained elsewhere in the architecture.

We study the marginal contribution of projector-side spatial PE in the public PS3 path of VILA-HD-8B [1]. The No-Projector-PE condition still retains upstream positional information: PS3 resamples the vision encoder’s positional embedding and gathers it using the patch-selection map before the projector. This motivates two controlled questions:

how much aggregate task performance is contributed by an *additional* projector spatial PE under this retained positional pathway, and which compact representation provides the best accuracy-parameter tradeoff when such a term is used? We compare no additional projector spatial PE, a learned grid, Cartesian Fourier features, selection-centered log-polar Fourier features (LogRetina), and a PoPE-inspired uniform-polar Fourier representation [4].

Our contributions are:

1. We audit the PS3 positional pathway and conduct a matched, three-seed comparison of five projector conditions plus a learned-grid initialization control. Their close aggregate results characterize the additional projector PE as a marginal component in this setting and establish retained upstream position as an essential control for interpreting projector-PE comparisons.
2. Among the tested explicit PEs, uniform-polar Fourier achieves the highest observed ten-benchmark mean, $77.020 \pm 0.072\%$, while using $5.65 \times$ fewer PE parameters than the learned grid. The zero-initialized grid obtains $76.899 \pm 0.181\%$; their +0.121-point mean difference changes sign across seeds, indicating comparable aggregate accuracy with substantially lower PE-module parameterization.
3. Holding the Cartesian PE fixed, we measure a 3.59-point OCR-5 change at the same evaluated step across planned cosine-schedule horizons. This effect substantially exceeds the matched PE differences and establishes scheduler matching as an essential control for low-budget projector-PE comparisons.

2. RELATED WORK

Position in high-resolution VLMs. M-RoPE [6] decomposes position across temporal, height, and width dimensions and is used by Qwen2.5-VL [5]; V2PE [10] studies smaller, variable visual-token position increments for multimodal long contexts. PyPE [29] instead changes the ordering of visual position indices to mitigate long-range decay in LLM-side

RoPE. SLiME [3] and InternVL [2] provide high-resolution visual processing strategies for multimodal models. These methods operate on backbone or LLM-side positions; we isolate the marginal contribution of an additive spatial term at the projector while preserving the rest of the PS3 pathway.

Spatial prompts and projector design. PIN [30] learns a small spatial prompt inside a frozen VLM to enable zero-shot object localization. Honeybee [31] redesigns the projector to preserve locality while controlling the number of visual tokens. The public PS3/VILA-HD implementation [1], by contrast, combines prompt-aware selection with a learned $729 \times C$ spatial grid at the projector. We neither introduce localization supervision nor replace the projector/token sampler; we ablate and replace only this additional PS3 spatial term.

Functional PE and necessity studies. Perceiver-VL [32] reports comparable performance with learned and Fourier PE for its latent array, providing a precedent for closely matched PE configurations in a different architecture. Functional PE generates position from coordinates and avoids a resolution-indexed lookup table. Fourier mappings follow random features [8] and neural coordinate encodings [7]; PoPE [4] instead decouples content and position in RoPE-based sequence models and is not a 2-D VLM-projector method. FlexiViT [27], NaViT [28], and RoPE-ViT [9] motivate spatial representations that remain usable across varying patch sizes, image resolutions, or aspect ratios. Our distinct focus is a matched comparison among No-PE, learned, and functional configurations after auditing position already retained upstream.

3. PROJECTOR-SIDE POSITION REPRESENTATIONS

For normalized global coordinates $(x, y) \in [-1, 1]^2$, we test: (i) no additional projector spatial PE; (ii) a learned grid $P_{\text{grid}} \in \mathbb{R}^{729 \times C}$; (iii) Cartesian Fourier PE; (iv) selection-centered log-polar Fourier PE (LogRetina); and (v) uniform-polar Fourier PE. The functional variants first map coordinates to (u, v) and then apply the same fixed Fourier basis and learned projection:

$$\phi(u, v) = [\sin(2\pi 2^k u), \cos(2\pi 2^k u), \sin(2\pi 2^k v), \cos(2\pi 2^k v)]_{k=0}^{K-1}, \quad (1)$$

$$e(x, y) = W\phi(u, v) + b, \quad W \in \mathbb{R}^{C \times 4K}. \quad (2)$$

Cartesian uses $(u, v) = (x, y)$. Uniform polar uses normalized radius and angle. LogRetina centers coordinates at the weighted selection centroid $(\bar{x}, \bar{y})$ and uses

$$r' = \frac{\log(1 + \alpha r)}{\log(1 + \alpha)}, \quad (u, v) = (r' \cos \theta, r' \sin \theta), \quad (3)$$

with $\alpha = 1$. With $K = 32$ and $C = 3584$, every functional variant has $(4K + 1)C = 462,336$ learned parameters, versus

$729C = 2,612,736$ for the grid: a $5.65\times$ PE-module reduction. Its parameter count is independent of the spatial grid, so changing grid size does not require interpolating a learned table. This reduction applies specifically to PE-module parameterization; end-to-end model size remains dominated by the 8B backbone. Initialization is part of each tested configuration: standard Learned-Grid PE is reinitialized from $\mathcal{N}(0, 0.02^2)$, whereas the new functional projection weights and biases are zero-initialized so that they initially preserve the pretrained-model function. We therefore also report a zero-initialized Learned-Grid control (A0), which uses the identical training and evaluation protocol. This diagnostic separates the initialization question from the five primary configurations without replacing the standard grid reference. A separate sparse diagnostic additionally evaluates Fixed-Origin Log-Polar Fourier PE (internal code D); it is not a primary matched-table row.

4. EXPERIMENTS

4.1. Setup

All comparisons use the same VILA-HD-8B [1] PS3-1.5K Stage-2 checkpoint, with a Qwen2-family 7B language backbone and a SigLIP2 vision tower [13]. The ten benchmarks comprise five OCR tasks—TextVQA [18], DocVQA [19], ChartQA [20], OCRBench [21], and InfoVQA [17]—hereafter denoted **OCR-5**; four spatial/grounding tasks—V*Bench [23], MMBench [25], GQA [26], and POPE [24]—and MathVista [22]. OCRBench uses the repository’s normalized STVQA-style accuracy for within-study comparison rather than an official leaderboard score. TextVQA uses its validation split, MathVista uses `testmini`, and GQA uses balanced test-dev. DocVQA and InfoVQA use ANLS; ChartQA uses the repository’s normalized exact/numeric-tolerance accuracy, and the remaining tasks use their repository accuracy evaluators. For each seed, OCR-5 and 10-benchmark results are unweighted macro-averages of the corresponding percentage scores.

The 81k SFT mixture contains 12k each from ChartQA and DocVQA, 25k from ShareGPT4V [14], 24k from SVIT-Conv/CR [15], and 8k from Tulu [16]. All runs use effective batch size 64, LoRA [11] rank $r=64$ and scaling $\alpha=128$, 4-bit QLoRA [12], and one RTX 4090. A full-budget one-epoch run takes approximately 43 h; the matched 312-step runs take 10.6–10.7 h according to their trainer states.

The primary comparison uses independent 312-step runs with the same data mixture and seeded-sampler protocol, effective batch size 64, cosine horizon, and seeds 42/1234/2024. Approximately one quarter of an epoch is observed; this is a compute-budget condition, not a fixed 25% subset. We use formal method names in the paper; internal experiment codes are N (No Projector Spatial PE), A (Learned-Grid PE), A0 (zero-initialized Learned-Grid con-

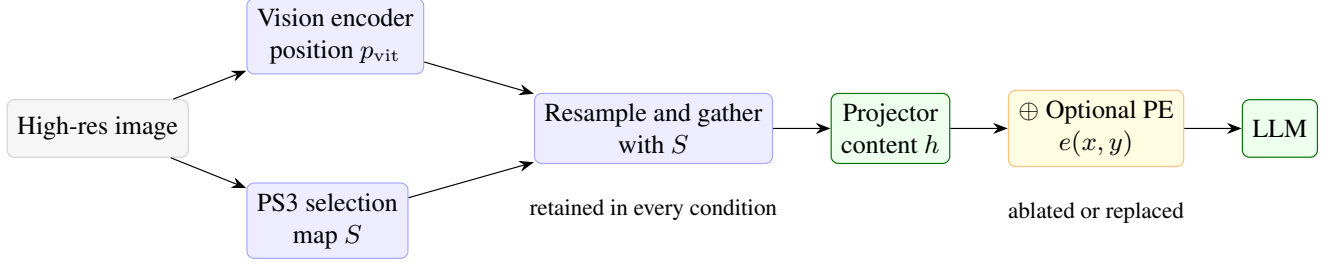


Fig. 1. Audited positional pathway. PS3 resamples vision-encoder position embeddings and gathers them with the selection map before the projector. Our no-projector-PE condition removes only the optional spatial term $e(x, y)$; upstream position and PS3 scale embeddings remain unchanged.

Table 1. Matched 312-step results over three seeds (%). A0 is an initialization diagnostic; all rows share exposure, horizon, and evaluation.

Method	PE Params	OCR-5	10-benchmark
No Projector Spatial PE	0	77.020±0.105	76.879±0.323
Learned-Grid PE	2.61M	76.570±0.352	76.847±0.123
Learned-Grid PE, zero init (A0)	2.61M	76.833±0.197	76.899±0.181
Cartesian Fourier PE	0.46M	76.696±0.044	76.893±0.128
Selection-Centered Log-Polar PE	0.46M	76.759±0.254	76.901±0.150
Uniform-Polar Fourier PE	0.46M	77.078±0.140	77.020±0.072

trol), C (Cartesian Fourier PE), E (Selection-Centered Log-Polar Fourier PE), and F (Uniform-Polar Fourier PE). The N condition zeros only the additional spatial PE module; vision-encoder position and PS3 high/low-resolution scale embeddings are retained.

Statistical scope: All five primary rows and the A0 diagnostic use matched exposure, schedule, deterministic decoding, and token limits. Standard deviations are sample SDs over the three predetermined seed-level macro-averages. With $n = 3$, we report observed means and paired differences rather than population-level significance. The archived Cartesian step-312 checkpoint from a 1263-step schedule appears only in the separate scheduler control.

4.2. Main Results

Table 1 answers the necessity question first. No Projector Spatial PE is within 0.032 points of Learned-Grid PE and within 0.141 points of the best mean over ten benchmarks; its paired ordering against Uniform-Polar Fourier reverses across seeds. The tested system therefore characterizes the additional projector spatial PE as a marginal component under the matched setting, while the upstream positional pathway in Fig. 1 remains active.

Among explicit PEs, Uniform-Polar Fourier has the highest mean and lowest observed across-seed dispersion of the ten-benchmark macro-average. A0 differs from the standard grid by only +0.052 points on average, with mixed paired signs. Uniform-Polar differs from A0 by +0.251, +0.284, and

Table 2. Same Cartesian Fourier PE evaluated at step 312 under different cosine horizons; OCR-5 over three seeds (%).

Condition	Seed scores	Mean	σ
Step 312 / horizon 1263	72.49/73.99/72.84	73.11	0.78
Step 312 / horizon 312	76.70/76.74/76.65	76.70	0.04

−0.171 points for seeds 42, 1234, and 2024 (mean +0.121); the mixed paired signs characterize the aggregate difference as small relative to seed variation. Cartesian, LogRetina, and Uniform-Polar remain close, while all functional variants use $5.65\times$ fewer PE parameters than the learned grid.

4.3. Scheduler-Horizon Control

The archived run and matched run use the same data mixture, seeded-sampler protocol, model family, Cartesian PE, optimizer hyperparameters, evaluated step, and seeds; the treatment is the planned schedule horizon. Because the warmup ratio remains 0.03, the absolute warmup also changes from 38 to 10 steps together with the cosine trajectory. The logged step-312 learning rates are 1.763×10^{-5} and zero. Changing the complete learning-rate schedule raises the mean by 3.59 points, much more than any primary-table difference. A training fraction must therefore specify both data exposure and scheduler horizon; an intermediate checkpoint of a long schedule is not equivalent to a completed short-budget run.

4.4. Interpretation and Scope

The close means indicate that coordinate-system choice has only a small aggregate effect under the matched setting. LogRetina also changes the origin using the selection centroid, so its comparison with fixed-origin Cartesian and uniform-polar encodings reflects complete configurations rather than an isolated geometry intervention. In a separate one-round, single-seed diagnostic, fixed- and dynamic-origin log-polar variants obtain 75.8% and 75.9%, respectively, placing the observed centroid effect within the narrow range seen among the primary coordinate variants. Log compression therefore

remains a foveated design choice whose accuracy benefit may depend on broader backbone, resolution, or training regimes.

Scope and limitations. The controlled evaluation covers one VILA-HD-8B backbone and three seeds, while the centroid diagnostic uses a single seed. The demonstrated $5.65\times$ reduction applies to PE-module parameterization, which is small relative to the 8B model, and does not establish end-to-end compression or latency gains. Multi-backbone and cross-resolution evaluation provides a natural next step for testing the generality of these findings.

5. CONCLUSION

We audit the marginal contribution of projector-side spatial PE in high-resolution VILA-HD-8B while retaining its upstream selection-aware positional pathway. Across five primary conditions and an initialization control, the three-seed ten-benchmark means span only 0.173 points, characterizing the additional projector PE as a marginal component in this setting. Among the tested explicit PEs, Uniform-Polar Fourier achieves the highest observed mean while using $5.65\times$ fewer PE parameters than the grid controls and avoiding grid-table interpolation. Its mixed seed-level ordering relative to the initialization-matched grid further characterizes their aggregate accuracy as comparable.

The larger finding is methodological: changing the planned cosine-schedule horizon moves the same Cartesian PE by 3.59 OCR-5 points, substantially more than the matched differences among PE configurations. Reliable projector-PE comparisons therefore require an explicit audit of retained upstream positional information together with matched data exposure and learning-rate scheduling. Under these controls, architectural context and training protocol exert larger observed effects than the tested coordinate refinements.

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